

Attuning to computational creativity: A socio-technical imaginary

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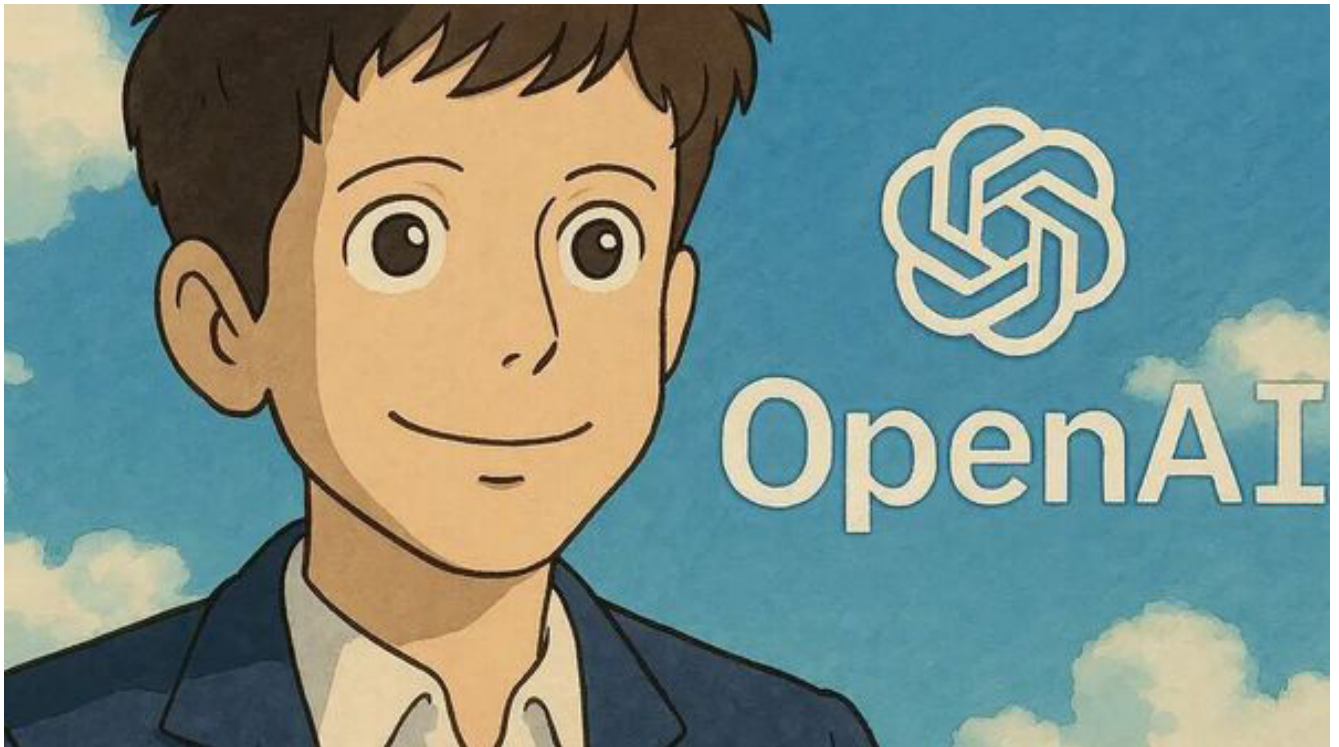
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ABSTRACT Louise Amoore and colleagues (2024) recently wrote that the generativity of GenAI systems extends beyond the creation of a ‘novel’ output, and into the creation of a new logic/ politics of AI. This paper proposes to build on this theoretical foundation to illustrate how GenAI systems forge an emerging aesthetics that arises from human-machine interaction. Our positioning of aesthetics is largely phenomenological – as both sensing and sense-making (Fuller and Weizman, 2021). We put forward ‘attuning’ as a necessary mode of perceiving oneself as being in creative partnership with AI tools. Building on Anna Munster’s recent work *DeepAesthetics* (2025), we argue that the necessity of attuning stems from the recognition of GenAI systems as not only technological, but also culturally constructed as well as *culturally constructing*. This socio-technical ensemble exceeds human-centred imaginaries. We argue for a recognition of the creative agency of AI models, alongside their material specificity. Through a discourse analysis of publicly available material from OpenAI on Sora and Sora 2 Pro, we identify three key terms: seamlessness, consistency, and simulation. We find that each term is represented by OpenAI as a ‘neutral’ technological affordance. Our analysis results in a counter perspective, finding that these processes are embedded with human-centred imaginaries from development through to interaction and content circulation. To address this disjunction, we put forward ‘attunement’ as mode of engagement. This enables users to perceive the entangled social, cultural, and technical layers that inform the aesthetics of AI and, consequently, building an understanding of the creative capacity of GenAI. Our paper makes two propositions. First, we consider creative agency through the user-model interaction and questioning the parameters of creative partnership. Second, it positions the generativity of models like Sora not as output, but as emerging aesthetic processes.



Introduction

In late 2025, OpenAI CEO Sam Altman released a blogpost about the company's latest text-to-video diffusion model, Sora 2. He claimed that, because of the model, "Creativity could be about to go through a Cambrian explosion." A year earlier, Louise Amoore and her collaborators (2024) wrote that Generative AI (GenAI) should not only be seen as generative in terms of content outputs, but as generative of world logic(s). The technical configuration of GenAI – in particular, the computational logics of latent spaces – constructs the parameters of a model's 'perspective'. Through increasingly automated forms of decision-making, the model shapes actions and consequences in everyday life. And yet, humans are invariably present in relation to machines and models. It is worth drawing attention to Anna Munster (2025), who put forward that the aesthetics of GenAI should be understood through 'deepaesthetics,' a recognition

of the socio-technical entanglement of sensing and sense-making in relations with GenAI. Positioning both Amoore et al.'s (2024) focus on the socio-political ramifications of this trajectory and Munster's (2025) 'deepaesthetics' in relation to Altman's claims, lays the foundation to justify a closer look at how GenAI models, as world models, configure perceptions of creative agency.

We posit that understanding AI as a creative collaborator requires attunement. Attunement in this instance is about locating points of affective resonance experienced by humans in relation to machines. These resonances do not emerge in a vacuum. They are in part configured by how technologies are coded – socially, culturally, and technically. This multi-layered encoding is often blurry. It is difficult to recognize what features of the AI model stem from culturally embedded finetuning, and which are computational constraints. Attuning to the deepaesthetics emerging with text-to-video requires a balance between computational

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[1] - Specifically, Sora relies on diffusion models and transformers. A more in-depth technical explanation for Sora as a diffusion transformer is offered by Peebles, Brooks et al. (2024), and for diffusion models more generally, see Sunde (2025).

operations, and how we experience and explain these processes through our human logics.

Computation operates on a logico-mathematical structure that is often in tension with human emotional resonances. This is important on two levels. First, it means that our experiences are often shaped by our familiarity with other technologies that have little to do with the AI model in question. Second, it enables the AI industry to push narratives about what and how the model ‘should’ be used and understood. Understood through a critical lens, the AI industry ‘hijacks’ the narrative about how to relate to the technology and thus informally ‘trains’ the user to attune in a manner which is beneficial to the company producing the technology. Rather than exploring deepaesthetics, this use reiterates existing aesthetic regimes stemming from cultural and/or social expectation. To illustrate this phenomenon, this paper analyses the technical operation and the marketing rhetoric surrounding OpenAI’s text-to-video model Sora.

Introducing Sora

OpenAI publicly launched Sora at the end of 2024. The initial iteration of OpenAI’s text-to-video technology is geared towards generating what OpenAI calls ‘high-fidelity’ short videos from text prompts (Brooks, Peebles, et al., 2024). Since then, OpenAI has divided its text-to-video models into two categories – Sora 2 for the everyday user, and Sora 2 Pro for the creative professional. While the former claims to prioritize turnaround over fidelity, the latter maintains ‘high-resolution cinematic footage’ and ‘visual precision’ as cornerstones of the model (OpenAI API, n.d.). The conflation of high-resolution and visual precision is not new to text-to-image cultures. Because resolution can be quantified, the developers of visual technologies have long chased higher-resolution, using it as a sign of visual quality. By emphasizing high-resolution, OpenAI is placing Sora within an ideology that Francesco Casetti and Antonio Somaini have identified as one, “that weaves together values such as mimetic precision, sensory enhancement, immersive participation, technical performance, and a view of technological progress” (Casetti & Somaini, 2018, 2).

This ideology is commonly embedded through promotional and marketing discourse.

The so-called “generalist capabilities” of Sora are achieved by training the model on what OpenAI describes as “internet-scale data.” (Peebles and Brooks, n.d.). The scale at which GenAI operates enables these models to respond to a variety of prompts without having been trained specifically on that scenario. Importantly, the output must comply with the user expectation. The model is therefore also trained to create the ‘correct’ associations between words and imagery. This is done through tokens. In place of the text token common in Large Language Models (LLMs), Sora relies on visual patches (Peebles and Brooks, n.d.). These visual patches are compressed into Sora’s latent space by reducing the temporal and spatial dimensionality into latent representations (whereby mathematical representations stand in for the visual representations familiar to human vision). It is within this latent space that Sora both trains and generates its videos. To transform these outputs into content recognizable to human aesthetics, Sora is kitted with a decoder transforming the information into pixels. This architecture enables Sora to train efficiently and to be exposed to visual difference (for instance, duration, resolution, ratio, and grid-size), and to produce outputs that mimic familiar media formats (camera angles, subject consistency, visual styles, etc.) [1].

Aims and objectives

Existing user studies of Sora tend to focus on people’s concerns and perceptions of impact (Zhou et al. 2024), or the model’s technical limitations and opportunities (Liu et al. 2024). Our objective in this paper is to bring forward Open AI’s narrative around Sora and, in doing so, illuminate how AI models are formed through socio-cultural imaginaries. Attuning to the construction of cultural meaning around AI models, offers new insight into how AI models are discursively formed. Such techniques shape subsequent human-machine interactions, influencing the parameters of creative practices. By attending to key

discursive terms, we show that an understanding of the creative agency of AI models begins prior to interactions at the interface level. Our aim is to provide an approach for analysing a wide range of commercially available AI models.

Methods

The basis of this research is a discourse analysis of the publicly available material posted by OpenAI about Sora and Sora 2 Pro. This consisted of two official introductions to Sora, one white paper, and one promotional website. OpenAI CEO, Sam Altman, followed up with a blog post addressing the reception of the model, focusing on creative practices. From a close reading of this material, we located several key terms that emerged as foundational to the Sora discourse. We focused on terms that highlighted unspoken assumption, that implicitly evoked neutrality to ideologically frame the technology. Each term thus addresses a technical operation, whilst also positioning it within existing social constructs. Discourse analysis emphasizes the relationship between the discursive strategies employed by OpenAI and the techno-cultural experience they aim to engender. To further situate our analysis, we draw on some key concepts from existing theory – namely ‘frictionlessness’ (Kemper and Jankowski, 2024), ‘immediacy’ (Bolter and Grusin, 2000), and ‘deepaesthetics’ (Munster, 2025).

This is not intended to be a comprehensive account. Its scope is limited to a small selection of documents published by OpenAI employees and for the sake of brevity, has focused on the concepts associated with the located theme of ‘fidelity’. At the time of writing, Sora dominated the text-to-video landscape. It has since been discontinued. Future research may locate other, equally crucial, points of discursive meaning-making surrounding similar commercial AI models.

Results

The roll out of Sora was purportedly ‘slow’, and OpenAI began by granting artists, designers and filmmakers early access. This was done, according to

OpenAI, to learn how to make the model “most helpful for creative professionals.” (OpenAI: Sora, n.d.). OpenAI goes on to state that Sora will help creatives “generate complex scenes with multiple characters, specific types of motion, and accurate details of the subject and background.” (Fig. 1). The model can do so because it “understands not only what the user has asked for in the prompt, but also how those things exist in the physical world” (OpenAI: Sora, n.d.). On the same webpage, OpenAI attributes these capabilities to Sora’s “deep understanding of language,” allowing the model to create worlds and characters that “express vibrant emotions.” These statements are representative of the promotional and scientific introductions to Sora. They are also emblematic of the overall discourse around visual AI models and hold significance for the ways in which creativity is positioned by the GenAI industry. The discourse surrounding Sora and Sora 2 Pro positions the AI systems as ‘tools’ for creative support, rather than as a creative collaborator. With these systems, the creative professional can create audiovisual material that appears akin to ‘the real world’.

Pei-Sze Chow and Claudio Celis Bueno have critiqued the AI industry’s claim that new models, such as Sora, enhance human creativity. As they write, “[t]he premise here is that generative AI can now enhance, unlock, and turbo-charge one’s creativity with great ease, revolutionising and democratising the production of audiovisual content” (Chow and Celis Bueno, 2025, 2). Similar to our stance, Chow and Celis Bueno argue that the discourse around AI and creativity as promoted by the industry is that the tool is both ‘advancing’ human creativity while also taking on the role of a human ‘servile’ assistant. For Nataliia Laba (2025), it is the positioning of AI as a tool in the first place that is problematic. Laba writes that “the tool metaphor is a powerful rhetorical strategy” that lends “intellectual credibility to the industry’s preferred narrative” and ultimately avoids critical interrogation of AI as a value-laden, ideologically powerful technology. The internalization of the term ‘tool’ neutralizes AI to its use-value, passing over the politics of generativity and world logics. It positions AI as a

[2] - OpenAI touches on notions of positivism with this statement around purpose simulators, and disregard the complex (and well researched) history of indexicality and realism in visual culture.

creative support, without recognizing its capacity for creative agency. Our analysis of Sora highlights how the socio-technical imagination of AI plays out through the concept of fidelity, and its associated terms of consistency, seamlessness, and simulation.

Fidelity

OpenAI promotes a discourse that positions this concept of ‘fidelity’ as inherent to the model. Fidelity is upheld through seamlessness in use and visuality, consistency in transitions, output, and remediation, and reference to ‘real world’ simulation. These terms are socially coded, as well as technologically encoded.

Fidelity is a commonly used term in OpenAI’s research. While it is never explicitly defined by OpenAI’s researchers, it is used in a context and representational manner, such that it can be understood to mean photorealism (Sunde, 2025). OpenAI continues to conflate ‘fidelity’ with the way in which we have grown accustomed to the physical world being mediated through technology. In other words, ‘photorealism’ in the context of OpenAI means that the image appears to have been produced using optical technology *and* (crucially) that this means the image

is ‘true’ (holds fidelity) to the real world. For OpenAI, the socio-technical optimisation towards realism has an additional function. OpenAI describes this desire to generate high-fidelity video as “a promising path towards building general purpose simulators of the physical world.” (Brooks, Peebles, et al., 2024). Therefore, the desire to create videos that appear to be “records” of the physical world is to create a model that can function as an indexical twin. This is based on an unstable assumption that the world is knowable, capturable, and reproducible [2]

Seamlessness

Brooks, Peebles et al. (2024) note that Sora is “capable of extending videos, either backwards or forwards in time” which can “produce a seamless infinite loop.” They go on to write that “We can also use Sora to gradually interpolate between two input videos, creating seamless transitions between videos with entirely different subjects and scene compositions.” ‘Seamlessness’ invokes both a technical operation and an ideal user experience. This is reminiscent of existing work on the concept of ‘frictionlessness.’ Jakko Kemper and Steve Jankowski offer insight into the alignment between the

Fig. 1 - Screenshot of moving image and accompanying text prompt from OpenAI Sora product release. Source: OpenAI. <https://openai.com/index/sora-2/>. (2025)



Prompt: a person is standing on 2 horses with legs spread. make it not slowmo also realistic. the guy fell off pretty hard in the end. single shot



Prompt: Two mountain explorers in bright technical shells, ice crusted faces, eyes narrowed with urgency shout in the snow, one at a time

design philosophy of frictionlessness, and the ‘ideological and economic interests’ of Silicon Valley (2024, 287). They write that the term implies a seamlessness of technological embedding into everyday life (2024, 287-88).

In the case of Sora, OpenAI positions seamlessness as realized through the visual output matching the expectation of the user as they prompt the model. Statements like “the model is able to follow the user’s text instructions in the generated video more faithfully” (OpenAI: Sora) are indicative of a desire to eliminate disruptive user experiences in text-to-video generation. However, it may be that this directly contradicts collaborative creative processes, which are often stimulated by friction. Tomas Lawton and colleagues (2023) found in their research that users of AI systems felt they benefitted creatively when they experience moments of disruption. As Lawton et al write, this contrasts “with traditional notions of creativity support that focus on maximally unobtrusive tools.” (2023, 1995). This shows a disconnect between the corporate discourse and technical orientation towards frictionlessness, and user attunement towards creativity.

Consistency

There are two interrelated considerations in unpacking the term consistency as invoked by OpenAI. The first is through the notion of spatial and temporal consistency via things like character persistence. The second is the notion of experiential consistency, whereby the visual appearance of the output aligns with our expectations of existing mediums [3] (Fig. 2). Brooks, Peebles et al. (2024) write that “Sora can generate videos with dynamic camera motion. As the camera shifts and rotates, people and scene elements move consistently through three-dimensional space.” (Brooks, Peebles, et al.). Sora does not have a camera, but nonetheless, OpenAI claim it has ‘dynamic camera motion.’ Similarly, on the OpenAI API Guide to Sora, creativity is directly linked with prompting visual cues associated with cinematography: “The prompt defines the creative look and feel – subjects, camera, lighting, and motion.” These visual elements construct a world of familiarity with the viewer, placing the video within a frame of reference. The deliberate use of such signals is also the reason why the experience and technicity of GenAI often diverge.

Attuning to the socio-technics of Sora and Sora 2 Pro positions the creative

Fig. 2 - Screenshot of moving image and accompanying text prompt from OpenAI Sora product release. Source: OpenAI. <https://openai.com/index/sora-2/>. (2025)

[3] - We understand this to follow the aesthetic logic of immediacy (Bolter and Grusin 2000).

professional to work with both the processual and output-oriented aesthetics of the model. This, in turn, encourages recognition of the model as an AI *system* rather than an AI tool. As of yet, OpenAI ‘consistently’ emphasizes the output of Sora, while treating the computational operations as a neutral mechanism to achieve content generation. This aligns with the well-established concept of ‘immediacy’ put forward by Jay David Bolter and Richard Grusin over a quarter of a century ago. They write that “immediacy dictates that the medium itself should disappear and leave us in the presence of the thing represented” (Bolter and Grusin 2000, 6). When introducing immediacy at the turn of the century, it was explicitly dealing with the digital turn. This moment was reliant on human control for the generation of content – renders were still hand-drawn, albeit digitised. GenAI turns the dial further – both the medium and the thing being represented are one and the same, both are made to disappear, we are left in the simulacra. In more practical terms, bypassing the technical operations, OpenAI overlooks one of the key creative benefits for human-machine collaboration – the capacity to attune to computational aesthetics and extend, rather than replicate, human perspective.

Simulation

Fidelity, seamlessness, and consistency lead us to what OpenAI purports to be the core of Sora’s potential: its capacity for simulation. They write that: “We believe the capabilities Sora has today demonstrate that continued scaling of video models is a promising path towards the development of capable simulators of the physical and digital world, and the objects, animals and people that live within them.” (Brooks, Peebles, et al., 2024). However, as we have put forward in our analysis, Sora is optimised for a specific form of realism that mimics existing creative mediums. As such, simulation is not designed to stimulate a new form of creative practice, but to remediate existing visual cultures. By referencing both physical *and* digital simulation, OpenAI acknowledge that what Sora replicates is not necessarily the physical world per se, but the existing means used to replicate the physical world. Building these simulators, OpenAI ar-

gues, is “an important milestone for achieving AGI” (Brooks, Peebles, et al 2024). We have shown this to be a flawed premise, particularly as Sora is learning to simulate the physical world only through the data being recorded. The representations and spatializations are constrained within a visual genealogy that adheres to the ideology of realism taken up by OpenAI. Representationalism affords a simpler way for humans to make sense of computational aesthetics. However, as Munster (2025) shows through *deepaesthetics*, this is a highly limited approach to cultural meaning-making.

Conclusion

In many respects, this paper has been a response to the wager put forward by Munster in *DeepAesthetics* (2025). She writes: “My wager will be that by bringing careful, granular attention to ML’s processuality – that domain of computational experience registering yet often beyond perception – we can scope out what is singular about the sensibility engendered by its aesthetic individuation(s)” (Munster, 2025, 10). By conducting an analysis of the discourse that OpenAI has constructed around Sora, we located a contradiction that is significant for human-machine co-creation. Sora is presented as the next innovation that will help creative professionals in their work. OpenAI relies on a mythopoetic language, the ‘Cambrian explosion,’ in which the capabilities of Sora are suggested to have a kind of creative agency. At the same time, the claims of frictionlessness position this agency within a corporate model of control. The corporate standpoint thus seems that the creative professional does not have to concern themselves with the creative agency of the model, which is harnessed by OpenAI.

Claims around the model’s language capabilities and strong emotional resonances position the model as a transparent media that can be used in place of real-world interaction. At the same time, the media is at the disposal of the human user, designed to be a frictionless ‘tool’ that seamlessly operates according to the human user experiences. Yet there is incongruity between corporate

discourse and user attunement. OpenAI's language suggests that Sora has an embodied understanding of the videos it creates. At the same time, this 'creative' capacity is constrained to a narrow output, based on a niche understanding

of fidelity as a gateway to real-world interaction. In other words, the new 'tool' operates within a logic of aesthetics derived from cinematic technologies, and Sora's agency is constrained to operate within this aesthetic regime.

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